Automatic Debiased Machine Learning via Riesz Regression

Semiparametric Efficiency Reading Group

Houssam Zenati

March 2026

Outline

1. Motivation
2. Base setup
3. Theory
4. Extensions
5. Empirical message
6. Takeaways

Motivation

Automatic Debiased Machine Learning via Riesz Regression

Why this paper?

- Machine learning methods can estimate high-dimensional regressions well.

Why this paper?

- Machine learning methods can estimate high-dimensional regressions well.
- But naive plug-in estimation of a target functional inherits regularization bias.

Why this paper?

- Machine learning methods can estimate high-dimensional regressions well.
- But naive plug-in estimation of a target functional inherits regularization bias.
- Orthogonal scores and cross-fitting remove this first-order bias.

Why this paper?

- Machine learning methods can estimate high-dimensional regressions well.
- But naive plug-in estimation of a target functional inherits regularization bias.
- Orthogonal scores and cross-fitting remove this first-order bias.
- The missing ingredient is the debiasing function α_0.

Why this paper?

- Machine learning methods can estimate high-dimensional regressions well.
- But naive plug-in estimation of a target functional inherits regularization bias.
- Orthogonal scores and cross-fitting remove this first-order bias.
- The missing ingredient is the debiasing function α_0.
- The paper's main idea is to estimate α_0 automatically through a Riesz regression problem.

$$\text{Auto-DML} = \text{learn } \gamma_0 \; + \; \text{learn } \alpha_0 \; + \; \text{orthogonal plug-in}$$

Running example: the ATE

We fix one target from the start.

$$W = (Y, D, Z), \qquad X = (D, Z), \qquad \gamma_0(d, z) = \mathbb{E}[Y \mid D = d, Z = z].$$

Running example: the ATE

We fix one target from the start.

$$W = (Y, D, Z), \qquad X = (D, Z), \qquad \gamma_0(d, z) = \mathbb{E}[Y \mid D = d, Z = z].$$

Our target is

$$\theta_0 = \mathbb{E}[\gamma_0(1, Z) - \gamma_0(0, Z)].$$

Running example: the ATE

We fix one target from the start.

$$W = (Y, D, Z), \qquad X = (D, Z), \qquad \gamma_0(d, z) = \mathbb{E}[Y \mid D = d, Z = z].$$

Our target is

$$\theta_0 = \mathbb{E}[\gamma_0(1, Z) - \gamma_0(0, Z)].$$

Equivalently,

$$\theta_0 = \mathbb{E}[m(W, \gamma_0)], \qquad m(W, \gamma) = \gamma(1, Z) - \gamma(0, Z).$$

Running example: the ATE

We fix one target from the start.

$$W = (Y, D, Z), \qquad X = (D, Z), \qquad \gamma_0(d, z) = \mathbb{E}[Y \mid D = d, Z = z].$$

Our target is

$$\theta_0 = \mathbb{E}[\gamma_0(1, Z) - \gamma_0(0, Z)].$$

Equivalently,

$$\theta_0 = \mathbb{E}[m(W, \gamma_0)], \qquad m(W, \gamma) = \gamma(1, Z) - \gamma(0, Z).$$

Under unconfoundedness, this is the average treatment effect.

Base setup

Automatic Debiased Machine Learning via Riesz Regression

Linear functional of a conditional mean

Section 2 considers targets of the form

$$\theta_0 = \mathbb{E}[m(W, \gamma_0)], \qquad \gamma_0(x) = \mathbb{E}[Y \mid X = x],$$

where the map

$$\gamma \mapsto \mathbb{E}[m(W, \gamma)]$$

is linear and continuous.

Linear functional of a conditional mean

Section 2 considers targets of the form

$$\theta_0 = \mathbb{E}[m(W, \gamma_0)], \qquad \gamma_0(x) = \mathbb{E}[Y \mid X = x],$$

where the map

$$\gamma \mapsto \mathbb{E}[m(W, \gamma)]$$

is linear and continuous.
Then by Riesz representation, there exists $\alpha_0(X)$ such that

$$\mathbb{E}[m(W, \gamma)] = \mathbb{E}[\alpha_0(X)\gamma(X)] \qquad \text{for all } \gamma.$$

Linear functional of a conditional mean

Section 2 considers targets of the form

$$\theta_0 = \mathbb{E}[m(W, \gamma_0)], \qquad \gamma_0(x) = \mathbb{E}[Y \mid X = x],$$

where the map

$$\gamma \mapsto \mathbb{E}[m(W, \gamma)]$$

is linear and continuous.
Then by Riesz representation, there exists $\alpha_0(X)$ such that

$$\mathbb{E}[m(W, \gamma)] = \mathbb{E}[\alpha_0(X)\gamma(X)] \qquad \text{for all } \gamma.$$

So the target functional induces a representer α_0, and this is the debiasing function entering the orthogonal score.

Orthogonal score

Given (θ, γ, α), define

$$\psi(W; \theta, \gamma, \alpha) = m(W, \gamma) - \theta + \alpha(X)(Y - \gamma(X)).$$

Orthogonal score

Given (θ, γ, α), define

$$\psi(W; \theta, \gamma, \alpha) = m(W, \gamma) - \theta + \alpha(X)(Y - \gamma(X)).$$

At the truth,

$$\mathbb{E}[\psi(W; \theta_0, \gamma_0, \alpha_0)] = 0.$$

Orthogonal score

Given (θ, γ, α), define

$$\psi(W; \theta, \gamma, \alpha) = m(W, \gamma) - \theta + \alpha(X)(Y - \gamma(X)).$$

At the truth,

$$\mathbb{E}[\psi(W; \theta_0, \gamma_0, \alpha_0)] = 0.$$

In the base conditional mean model:

- the score is orthogonal around (γ_0, α_0);

Orthogonal score

Given (θ, γ, α), define

$$\psi(W; \theta, \gamma, \alpha) = m(W, \gamma) - \theta + \alpha(X)(Y - \gamma(X)).$$

At the truth,

$$\mathbb{E}[\psi(W; \theta_0, \gamma_0, \alpha_0)] = 0.$$

In the base conditional mean model:

- the score is orthogonal around (γ_0, α_0);
- it is also doubly robust.

Orthogonal score

Given (θ, γ, α), define

$$\psi(W; \theta, \gamma, \alpha) = m(W, \gamma) - \theta + \alpha(X)(Y - \gamma(X)).$$

At the truth,

$$\mathbb{E}[\psi(W; \theta_0, \gamma_0, \alpha_0)] = 0.$$

In the base conditional mean model:

- the score is orthogonal around (γ_0, α_0);
- it is also doubly robust.

So inference reduces to estimating both γ_0 and the representer α_0.

ATE example: standard DML vs Auto-DML

For the ATE,

$$\alpha_0(X) = \frac{D}{\pi_0(Z)} - \frac{1-D}{1-\pi_0(Z)}, \qquad \pi_0(Z) = \mathbb{P}(D = 1 \mid Z).$$

ATE example: standard DML vs Auto-DML

For the ATE,

$$\alpha_0(X) = \frac{D}{\pi_0(Z)} - \frac{1-D}{1-\pi_0(Z)}, \qquad \pi_0(Z) = \mathbb{P}(D = 1 \mid Z).$$

Hence standard DML uses the familiar AIPW score

$$\psi(W; \theta, \gamma, \pi) = \gamma(1, Z) - \gamma(0, Z) - \theta + \left(\frac{D}{\pi(Z)} - \frac{1-D}{1-\pi(Z)} \right) (Y - \gamma(D, Z)).$$

ATE example: standard DML vs Auto-DML

For the ATE,

$$\alpha_0(X) = \frac{D}{\pi_0(Z)} - \frac{1-D}{1-\pi_0(Z)}, \qquad \pi_0(Z) = \mathbb{P}(D = 1 \mid Z).$$

Hence standard DML uses the familiar AIPW score

$$\psi(W; \theta, \gamma, \pi) = \gamma(1, Z) - \gamma(0, Z) - \theta + \left(\frac{D}{\pi(Z)} - \frac{1-D}{1-\pi(Z)} \right) (Y - \gamma(D, Z)).$$

The issue is not the ATE itself.

ATE example: standard DML vs Auto-DML

For the ATE,

$$\alpha_0(X) = \frac{D}{\pi_0(Z)} - \frac{1-D}{1-\pi_0(Z)}, \qquad \pi_0(Z) = \mathbb{P}(D = 1 \mid Z).$$

Hence standard DML uses the familiar AIPW score

$$\psi(W; \theta, \gamma, \pi) = \gamma(1, Z) - \gamma(0, Z) - \theta + \left(\frac{D}{\pi(Z)} - \frac{1-D}{1-\pi(Z)} \right) (Y - \gamma(D, Z)).$$

The issue is not the ATE itself.
The issue is that for more general targets:

- the representer need not be available in closed form;
- even when it is, explicit inverse-weight formulas can be unstable.

Riesz regression derivation

The paper characterizes the representer variationally.
Start from

$$\alpha_0 = \arg\min_{\alpha} \mathbb{E}\big[(\alpha_0(X) - \alpha(X))^2\big].$$

Riesz regression derivation

The paper characterizes the representer variationally.
Start from

$$\alpha_0 = \arg\min_{\alpha} \mathbb{E}\big[(\alpha_0(X) - \alpha(X))^2\big].$$

Expanding the square gives

$$\alpha_0 = \arg\min_{\alpha} \left\{ -2\mathbb{E}[\alpha_0(X)\alpha(X)] + \mathbb{E}[\alpha(X)^2] \right\}.$$

Riesz regression derivation

The paper characterizes the representer variationally.
Start from

$$\alpha_0 = \arg\min_{\alpha} \mathbb{E}\big[(\alpha_0(X) - \alpha(X))^2\big].$$

Expanding the square gives

$$\alpha_0 = \arg\min_{\alpha} \left\{ -2\mathbb{E}[\alpha_0(X)\alpha(X)] + \mathbb{E}[\alpha(X)^2] \right\}.$$

By the Riesz identity,

$$\mathbb{E}[\alpha_0(X)\alpha(X)] = \mathbb{E}[m(W, \alpha)].$$

Riesz regression derivation

The paper characterizes the representer variationally.
Start from

$$\alpha_0 = \arg\min_{\alpha} \mathbb{E}\big[(\alpha_0(X) - \alpha(X))^2\big].$$

Expanding the square gives

$$\alpha_0 = \arg\min_{\alpha} \left\{ -2\mathbb{E}[\alpha_0(X)\alpha(X)] + \mathbb{E}[\alpha(X)^2] \right\}.$$

By the Riesz identity,

$$\mathbb{E}[\alpha_0(X)\alpha(X)] = \mathbb{E}[m(W, \alpha)].$$

Therefore

$$\alpha_0 = \arg\min_{\alpha} \left\{ -2\mathbb{E}[m(W, \alpha)] + \mathbb{E}[\alpha(X)^2] \right\}.$$

This is the population Riesz regression objective.

Cross-fitted Auto-DML estimator

Split the sample into folds. On each training fold:

- estimate $\hat{\gamma}_\ell$;

Cross-fitted Auto-DML estimator

Split the sample into folds. On each training fold:

- estimate $\hat{\gamma}_\ell$;
- estimate $\hat{\alpha}_\ell$ by empirical Riesz regression.

Cross-fitted Auto-DML estimator

Split the sample into folds. On each training fold:

- estimate $\hat{\gamma}_\ell$;
- estimate $\hat{\alpha}_\ell$ by empirical Riesz regression.

On the held-out fold, compute

$$\hat{\theta} = \frac{1}{n} \sum_{\ell=1}^{L} \sum_{i \in I_\ell} \left[m(W_i, \hat{\gamma}_\ell) + \hat{\alpha}_\ell(X_i)(Y_i - \hat{\gamma}_\ell(X_i)) \right].$$

Cross-fitted Auto-DML estimator

Split the sample into folds. On each training fold:

- estimate $\hat{\gamma}_\ell$;
- estimate $\hat{\alpha}_\ell$ by empirical Riesz regression.

On the held-out fold, compute

$$\hat{\theta} = \frac{1}{n} \sum_{\ell=1}^{L} \sum_{i \in I_\ell} \left[m(W_i, \hat{\gamma}_\ell) + \hat{\alpha}_\ell(X_i)\big(Y_i - \hat{\gamma}_\ell(X_i)\big) \right].$$

So the estimator keeps the usual one-step structure,

$$\text{plug-in term} + \text{orthogonal correction},$$

but learns the correction automatically.

Theory

Automatic Debiased Machine Learning via Riesz Regression

Theory: what must be shown?

There are two separate questions.

Theory: what must be shown?

There are two separate questions.
First:

Can we learn α_0 well from the Riesz regression problem?

Theory: what must be shown?

There are two separate questions.
First:

Can we learn α_0 well from the Riesz regression problem?

Second:

If $\hat{\alpha}$ and $\hat{\gamma}$ are accurate enough, do we get a CLT for $\hat{\theta}$?

Theory: what must be shown?

There are two separate questions.
First:

Can we learn α_0 well from the Riesz regression problem?

Second:

If $\hat{\alpha}$ and $\hat{\gamma}$ are accurate enough, do we get a CLT for $\hat{\theta}$?

So the theory has two layers:

- a statistical learning result for the representer;

Theory: what must be shown?

There are two separate questions.
First:

Can we learn α_0 well from the Riesz regression problem?

Second:

If $\hat{\alpha}$ and $\hat{\gamma}$ are accurate enough, do we get a CLT for $\hat{\theta}$?

So the theory has two layers:

- a statistical learning result for the representer;
- then a standard orthogonal-score asymptotic normality result for the target.

Main finite-sample result for the learned representer

Let

$$\alpha^\star = \arg\min_{\alpha \in \mathcal{A}} \left\{ -2\mathbb{E}[m(W, \alpha)] + \mathbb{E}[\alpha(X)^2] \right\}$$

be the best approximation to α_0 in the class $\mathcal{A}$.

Main finite-sample result for the learned representer

Let

$$\alpha^\star = \arg\min_{\alpha \in \mathcal{A}} \left\{ -2\mathbb{E}[m(W, \alpha)] + \mathbb{E}[\alpha(X)^2] \right\}$$

be the best approximation to α_0 in the class $\mathcal{A}$.
Under boundedness and complexity assumptions on the localized star hulls of

$$\mathcal{A} - \alpha_0 \qquad \text{and} \qquad m \circ \mathcal{A} - m \circ \alpha_0,$$

the paper proves that with high probability

$$\|\hat{\alpha} - \alpha_0\|^2 \lesssim M\delta_n^2 + \|\alpha^\star - \alpha_0\|^2 + \frac{M \log(1/\zeta)}{n}.$$

Main finite-sample result for the learned representer

Let

$$\alpha^\star = \arg\min_{\alpha \in \mathcal{A}} \left\{ -2\mathbb{E}[m(W, \alpha)] + \mathbb{E}[\alpha(X)^2] \right\}$$

be the best approximation to α_0 in the class $\mathcal{A}$.
Under boundedness and complexity assumptions on the localized star hulls of

$$\mathcal{A} - \alpha_0 \qquad \text{and} \qquad m \circ \mathcal{A} - m \circ \alpha_0,$$

the paper proves that with high probability

$$\|\hat{\alpha} - \alpha_0\|^2 \lesssim M\delta_n^2 + \|\alpha^\star - \alpha_0\|^2 + \frac{M \log(1/\zeta)}{n}.$$

Interpretation:

- δ_n is a critical-radius / localized-complexity term;

Main finite-sample result for the learned representer

Let

$$\alpha^\star = \arg\min_{\alpha \in \mathcal{A}} \left\{ -2\mathbb{E}[m(W, \alpha)] + \mathbb{E}[\alpha(X)^2] \right\}$$

be the best approximation to α_0 in the class $\mathcal{A}$.
Under boundedness and complexity assumptions on the localized star hulls of

$$\mathcal{A} - \alpha_0 \qquad \text{and} \qquad m \circ \mathcal{A} - m \circ \alpha_0,$$

the paper proves that with high probability

$$\|\hat{\alpha} - \alpha_0\|^2 \lesssim M\delta_n^2 + \|\alpha^\star - \alpha_0\|^2 + \frac{M \log(1/\zeta)}{n}.$$

Interpretation:

- δ_n is a critical-radius / localized-complexity term;
- $\|\alpha^\star - \alpha_0\|$ is the approximation error of the chosen class;

Main finite-sample result for the learned representer

Let

$$\alpha^\star = \arg\min_{\alpha \in \mathcal{A}} \left\{ -2\mathbb{E}[m(W, \alpha)] + \mathbb{E}[\alpha(X)^2] \right\}$$

be the best approximation to α_0 in the class $\mathcal{A}$.
Under boundedness and complexity assumptions on the localized star hulls of

$$\mathcal{A} - \alpha_0 \qquad \text{and} \qquad m \circ \mathcal{A} - m \circ \alpha_0,$$

the paper proves that with high probability

$$\|\hat{\alpha} - \alpha_0\|^2 \lesssim M\delta_n^2 + \|\alpha^\star - \alpha_0\|^2 + \frac{M \log(1/\zeta)}{n}.$$

Interpretation:

- δ_n is a critical-radius / localized-complexity term;
- $\|\alpha^\star - \alpha_0\|$ is the approximation error of the chosen class;
- this is the main new theorem in the paper.

Asymptotic normality of the final estimator

Once the nuisances are controlled, the orthogonal expansion gives

$$\hat{\theta} - \theta_0 = \frac{1}{n} \sum_{i=1}^{n} \psi(W_i; \theta_0, \gamma_0, \alpha_0) + o_p(n^{-1/2}).$$

Asymptotic normality of the final estimator

Once the nuisances are controlled, the orthogonal expansion gives

$$\hat{\theta} - \theta_0 = \frac{1}{n} \sum_{i=1}^{n} \psi(W_i; \theta_0, \gamma_0, \alpha_0) + o_p(n^{-1/2}).$$

The key rate condition is

$$\sqrt{n}\, \|\hat{\gamma} - \gamma_0\| \, \|\hat{\alpha} - \alpha_0\| \to 0.$$

Asymptotic normality of the final estimator

Once the nuisances are controlled, the orthogonal expansion gives

$$\hat{\theta} - \theta_0 = \frac{1}{n} \sum_{i=1}^{n} \psi(W_i; \theta_0, \gamma_0, \alpha_0) + o_p(n^{-1/2}).$$

The key rate condition is

$$\sqrt{n}\, \|\hat{\gamma} - \gamma_0\| \, \|\hat{\alpha} - \alpha_0\| \to 0.$$

Together with consistency of both nuisances, this implies

$$\sqrt{n}(\hat{\theta} - \theta_0) \implies N(0, V).$$

Asymptotic normality of the final estimator

Once the nuisances are controlled, the orthogonal expansion gives

$$\hat{\theta} - \theta_0 = \frac{1}{n} \sum_{i=1}^{n} \psi(W_i; \theta_0, \gamma_0, \alpha_0) + o_p(n^{-1/2}).$$

The key rate condition is

$$\sqrt{n}\, \|\hat{\gamma} - \gamma_0\| \, \|\hat{\alpha} - \alpha_0\| \to 0.$$

Together with consistency of both nuisances, this implies

$$\sqrt{n}(\hat{\theta} - \theta_0) \implies N(0, V).$$

So the CLT part is conceptually standard DML once the representer-learning theorem is in place.

What is actually new relative to standard DML?

Standard DML	**This paper**
Start from a hand-derived orthogonal score	Start from a target functional and learn the debiaser
Closed-form correction in simple cases	Generic variational characterization of the correction
Orthogonal expansion for $\hat{\theta}$	Orthogonal expansion for $\hat{\theta}$
Main burden: nuisance rates	Extra burden: learning theory for $\hat{\alpha}$

What is actually new relative to standard DML?

Standard DML	**This paper**
Start from a hand-derived orthogonal score	Start from a target functional and learn the debiaser
Closed-form correction in simple cases	Generic variational characterization of the correction
Orthogonal expansion for $\hat{\theta}$	Orthogonal expansion for $\hat{\theta}$
Main burden: nuisance rates	Extra burden: learning theory for $\hat{\alpha}$

So the new ingredient is not the final asymptotic linearity itself.

What is actually new relative to standard DML?

Standard DML	**This paper**
Start from a hand-derived orthogonal score	Start from a target functional and learn the debiaser
Closed-form correction in simple cases	Generic variational characterization of the correction
Orthogonal expansion for $\hat{\theta}$	Orthogonal expansion for $\hat{\theta}$
Main burden: nuisance rates	Extra burden: learning theory for $\hat{\alpha}$

So the new ingredient is not the final asymptotic linearity itself.
The new ingredient is the generic estimation and analysis of the Riesz representer.

Extensions

Automatic Debiased Machine Learning via Riesz Regression

Extension 1: generalized regressions

The paper next considers nuisances defined through moment restrictions,

$$\mathbb{E}[\rho(W, \gamma_0) b(X)] = 0 \qquad \forall b \in \Gamma.$$

Extension 1: generalized regressions

The paper next considers nuisances defined through moment restrictions,

$$\mathbb{E}[\rho(W, \gamma_0) b(X)] = 0 \qquad \forall b \in \Gamma.$$

This covers generalized linear and other M-estimation type regressions.

Extension 1: generalized regressions

The paper next considers nuisances defined through moment restrictions,

$$\mathbb{E}[\rho(W, \gamma_0) b(X)] = 0 \qquad \forall b \in \Gamma.$$

This covers generalized linear and other M-estimation type regressions.
The orthogonal score becomes

$$\psi(W; \theta, \gamma, \alpha) = m(W, \gamma) - \theta + \alpha(X)\rho(W, \gamma).$$

Extension 1: generalized regressions

The paper next considers nuisances defined through moment restrictions,

$$\mathbb{E}[\rho(W, \gamma_0) b(X)] = 0 \qquad \forall b \in \Gamma.$$

This covers generalized linear and other M-estimation type regressions.
The orthogonal score becomes

$$\psi(W; \theta, \gamma, \alpha) = m(W, \gamma) - \theta + \alpha(X)\rho(W, \gamma).$$

So the conditional-mean score is replaced by a more general residual map $\rho(W, \gamma)$.

Weighted Riesz regression

Define

$$v_\rho(X) = \frac{\partial}{\partial a}\mathbb{E}[\rho(W, \gamma_0 + a) \mid X]\Big|_{a=0}.$$

Weighted Riesz regression

Define

$$v_\rho(X) = \frac{\partial}{\partial a}\mathbb{E}[\rho(W, \gamma_0 + a) \mid X]\Big|_{a=0}.$$

Then the representer solves the weighted objective

$$\alpha_0 = \arg\min_{\alpha \in \Gamma} \left\{ -2\mathbb{E}[m(W, \alpha)] - \mathbb{E}[v_\rho(X)\alpha(X)^2] \right\}.$$

Weighted Riesz regression

Define

$$v_\rho(X) = \frac{\partial}{\partial a}\mathbb{E}[\rho(W, \gamma_0 + a) \mid X]\Big|_{a=0}.$$

Then the representer solves the weighted objective

$$\alpha_0 = \arg\min_{\alpha \in \Gamma} \left\{ -2\mathbb{E}[m(W, \alpha)] - \mathbb{E}[v_\rho(X)\alpha(X)^2] \right\}.$$

Relative to the base case, the quadratic curvature is now weighted by $v_\rho(X)$.

Weighted Riesz regression

Define

$$v_\rho(X) = \frac{\partial}{\partial a}\mathbb{E}[\rho(W, \gamma_0 + a) \mid X]\Big|_{a=0}.$$

Then the representer solves the weighted objective

$$\alpha_0 = \arg\min_{\alpha \in \Gamma} \left\{ -2\mathbb{E}[m(W, \alpha)] - \mathbb{E}[v_\rho(X)\alpha(X)^2] \right\}.$$

Relative to the base case, the quadratic curvature is now weighted by $v_\rho(X)$.
This is the technically central extension behind the more general theory.

Extension 2: nonlinear functionals of several nuisances

The paper also allows

$$\theta_0 = \mathbb{E}[m(W, \gamma_{0,1}, \ldots, \gamma_{0,J})],$$

with the target nonlinear in several nuisance functions.

Extension 2: nonlinear functionals of several nuisances

The paper also allows

$$\theta_0 = \mathbb{E}[m(W, \gamma_{0,1}, \ldots, \gamma_{0,J})],$$

with the target nonlinear in several nuisance functions.
The strategy is:

- differentiate m with respect to each nuisance direction;

Extension 2: nonlinear functionals of several nuisances

The paper also allows

$$\theta_0 = \mathbb{E}[m(W, \gamma_{0,1}, \ldots, \gamma_{0,J})],$$

with the target nonlinear in several nuisance functions.
The strategy is:

- differentiate m with respect to each nuisance direction;
- define one representer $\alpha_{0,j}$ per nuisance component;

Extension 2: nonlinear functionals of several nuisances

The paper also allows

$$\theta_0 = \mathbb{E}[m(W, \gamma_{0,1}, \ldots, \gamma_{0,J})],$$

with the target nonlinear in several nuisance functions.
The strategy is:

- differentiate m with respect to each nuisance direction;
- define one representer $\alpha_{0,j}$ per nuisance component;
- estimate each one from a derivative-based Riesz problem.

Extension 2: nonlinear functionals of several nuisances

The paper also allows

$$\theta_0 = \mathbb{E}[m(W, \gamma_{0,1}, \ldots, \gamma_{0,J})],$$

with the target nonlinear in several nuisance functions.
The strategy is:

- differentiate m with respect to each nuisance direction;
- define one representer $\alpha_{0,j}$ per nuisance component;
- estimate each one from a derivative-based Riesz problem.

So the basic architecture

$$\text{one nuisance} + \text{one representer}$$

extends to

$$\text{many nuisances} + \text{many representers}.$$

Empirical message

Automatic Debiased Machine Learning via Riesz Regression

What do the experiments show?

The empirical section mainly supports the methodological claim.

- Simulations show competitive performance.

What do the experiments show?

The empirical section mainly supports the methodological claim.

- Simulations show competitive performance.
- Direct representer learning can be more stable than explicit inverse-weight plug-ins.

What do the experiments show?

The empirical section mainly supports the methodological claim.

- Simulations show competitive performance.
- Direct representer learning can be more stable than explicit inverse-weight plug-ins.
- The mortgage denial application illustrates that the framework is usable on a real problem.

What do the experiments show?

The empirical section mainly supports the methodological claim.

- Simulations show competitive performance.
- Direct representer learning can be more stable than explicit inverse-weight plug-ins.
- The mortgage denial application illustrates that the framework is usable on a real problem.

This is primarily a method-and-theory paper; the experiments are there to show the approach is operational.

Takeaways

Automatic Debiased Machine Learning via Riesz Regression

Takeaways

- In the base setup, the paper learns the debiasing representer by solving a Riesz regression problem.

Takeaways

- In the base setup, the paper learns the debiasing representer by solving a Riesz regression problem.
- The generalized and nonlinear sections extend the same idea with weighted or derivative-based Riesz objectives.

Takeaways

- In the base setup, the paper learns the debiasing representer by solving a Riesz regression problem.
- The generalized and nonlinear sections extend the same idea with weighted or derivative-based Riesz objectives.
- The main technical novelty is the representer-learning theorem; the final CLT then follows by familiar orthogonal-score arguments.

$$\boxed{\text{Auto-DML} = \text{learn the nuisance} + \text{learn the representer} + \text{orthogonalize}}$$

Discussion questions

- When is direct representer learning preferable to plug-in learning of explicit inverse-weight ingredients?

Discussion questions

- When is direct representer learning preferable to plug-in learning of explicit inverse-weight ingredients?
- How far can this variational recipe be pushed in richer function spaces such as RKHS or operator-valued settings?

Discussion questions

- When is direct representer learning preferable to plug-in learning of explicit inverse-weight ingredients?
- How far can this variational recipe be pushed in richer function spaces such as RKHS or operator-valued settings?
- Can one systematically use it for more complex downstream causal or sequential targets?